Contents

Explainable Financial Fraud Detection Using Graph Neural Networks: A Comparative Study ..2

Abstract 2

Introduction 2

Literature Review 3

Graph Neural Networks for Financial Fraud Detection 3

GNNExplainer 4

PGExplainer 4

xFraud 5

SEFraud 5

Comparative Analysis and Research Motivation 6

Acknowledgment 7

References 7

Explainable Financial Fraud Detection Using Graph Neural Networks: A Comparative Study

Abstract

Financial fraud detection is difficult because fraudulent activities often involve complex relationships among transactions and financial entities. Traditional machine learning methods usually analyze transactions independently and may miss such relational patterns. Graph Neural Networks (GNNs) are suitable for this problem because they learn from both node features and graph structure. However, GNN predictions are often difficult to interpret, which is a major limitation in financial applications where decisions should be understandable.

This study focuses on explainable financial fraud detection using Graph Neural Networks through a comparative analysis of GNN models. GNNExplainer is used as a post-hoc explainability technique to identify important graph connections and node features responsible for individual predictions. The aim is to compare GNN-based fraud detection models while also examining how explainability can improve understanding of their decisions.

Introduction

Financial fraud has become increasingly complex with the rapid growth of digital transactions and online financial services. Traditional fraud detection methods often rely on individual transaction features, but many fraudulent activities involve relationships among multiple entities. Representing financial data as a graph allows such relationships to be analyzed directly.

Graph Neural Networks are designed to learn from graph-structured data by combining the features of a node with information from its neighboring nodes. This makes them suitable for detecting suspicious patterns in transaction networks. Recent studies have shown that GNNs can capture complex relational information that may be difficult to identify using conventional machine learning techniques.

Despite their effectiveness, GNNs are generally difficult to interpret. In fraud detection, simply predicting that a transaction or entity is fraudulent is not always sufficient; it is also important to understand which relationships and features influenced that prediction.

GNNExplainer was introduced to address this problem by identifying a compact subgraph and a subset of important node features that contribute to a GNN prediction.

Therefore, this research investigates **Explainable Financial Fraud Detection Using Graph Neural Networks: A Comparative Study**. The study aims to compare different GNN-based models for fraud detection and use GNNExplainer to interpret their predictions.

Literature Review

Graph-based machine learning has received increasing attention in financial fraud detection because many fraudulent activities involve relationships among multiple entities rather than isolated transactions. Existing research can broadly be divided into studies focusing on the application of Graph Neural Networks to fraud detection and studies focusing on explaining the predictions generated by graph-based models.

Graph Neural Networks for Financial Fraud Detection

Cheng et al. presented a comprehensive review of Graph Neural Networks for financial fraud detection. Their review examined more than one hundred studies and highlighted the ability of GNNs to capture complex relational patterns and dynamics within financial networks. The authors identified several major categories of GNN approaches used in fraud detection, including Graph Convolutional Networks, Graph Attention Networks, Graph Temporal Networks, and Heterogeneous Graph Neural Networks.

Graph Convolutional Networks aggregate information from neighbouring nodes to learn representations that incorporate both local attributes and graph connectivity. In financial transaction networks, this enables GCN-based approaches to capture cross-transaction patterns that may indicate fraudulent behaviour. However, GCNs can suffer from limitations such as over-smoothing and overfitting when the available labelled data is limited.

Graph Attention Networks extend neighbourhood aggregation by assigning different levels of importance to neighbouring nodes. Instead of treating all neighbours similarly, an attention mechanism allows the model to focus on the connections that are more relevant to a particular prediction. The financial fraud review notes that this adaptive weighting can improve the identification of complex fraudulent patterns compared with uniform neighbourhood aggregation.

Other GNN variants address additional characteristics of financial networks. Graph Temporal Networks incorporate temporal information to model changes in transaction behaviour over time, while Heterogeneous Graph Neural Networks support graphs containing multiple types of entities and relationships. These approaches are potentially useful in realistic financial systems where accounts, transactions, devices, customers, and other entities may interact through different relationship types. However, increased architectural complexity can also introduce higher computational requirements and scalability challenges.

The study by Cheng et al. therefore establishes that GNNs are highly relevant for financial fraud detection because of their ability to model relationships, automatically learn representations, and operate on complex transaction networks. At the same time, the authors identify scalability, adaptability, and interpretability as significant challenges requiring further research.

GNNExplainer

Ying et al. introduced **GNNExplainer** as one of the foundational techniques for explaining Graph Neural Network predictions. The method was proposed to address the difficulty of interpreting GNNs, whose predictions depend jointly on node attributes and relational graph structure.

GNNExplainer is a model-agnostic, post-hoc explanation technique. Given a trained GNN and a target prediction, the method identifies a compact subgraph of the original computation graph and a subset of node features that are most influential in producing the prediction. The approach can be applied to multiple types of graph-learning problems, including node classification, graph classification, and link prediction.

The central principle behind GNNExplainer is the maximization of mutual information between the GNN prediction and the generated explanation. The explanation is represented using learned graph and feature masks. Edges receiving higher mask values are interpreted as more important for the prediction, while the learned feature mask identifies influential feature dimensions.

This characteristic makes GNNExplainer particularly relevant for financial fraud detection. Instead of providing only a predicted class or fraud probability, the model can be examined to determine which neighbouring relationships and attributes influenced a particular decision. This can help reveal whether a fraud prediction is associated with meaningful graph patterns.

However, GNNExplainer primarily generates explanations by optimizing the explanation for individual instances. This can become computationally expensive when explanations are required for a large number of transactions or nodes.

PGExplainer

Luo et al. proposed **PGExplainer**, a parameterized explanation framework intended to address some limitations of instance-wise explanation methods. The authors observed that GNNExplainer generates customized explanations independently for individual predictions and therefore requires repeated optimization when many instances must be explained.

PGExplainer instead uses a neural network to parameterize the generation of explanations. The model learns distributions over graph edges and shares the parameters of the explanation network across multiple instances. As a result, once trained, PGExplainer can generate explanations for previously unseen instances without retraining the explanation process for each individual case.

The authors reported that PGExplainer produced competitive explanation quality while providing significant improvements in efficiency compared with GNNExplainer. However, PGExplainer focuses mainly on graph structural explanations through important edges,

whereas GNNExplainer can provide information regarding both graph structure and node features.

The comparison between these approaches illustrates an important trade-off in graph explainability: instance-specific optimization can provide detailed explanations, while parameterized approaches can improve scalability when a large number of predictions must be interpreted.

xFraud

Rao et al. proposed **xFraud**, an explainable fraud transaction detection framework designed for a real-world e-commerce environment. Unlike general-purpose GNN explanation studies, xFraud directly integrates graph-based detection and explainability within a financial transaction fraud scenario.

The xFraud framework consists primarily of a fraud detector and an explainer. The detector constructs a heterogeneous transaction graph containing multiple entity types, including transactions, payment tokens, email addresses, shipping addresses, and buyers. A heterogeneous GNN is then used to learn representations of these entities and predict whether incoming transactions are legitimate or fraudulent.

The explainer component is designed to provide understandable information about the predictions produced by the detector. xFraud combines information generated by GNNExplainer with graph-centrality measures to create a hybrid explanation mechanism. The authors also compared machine-generated explanations with human expert judgments to evaluate how closely the identified important graph components corresponded to expert reasoning.

One of the most important contributions of xFraud is its demonstration that explainable GNN-based fraud detection can operate on very large financial graphs. The framework was evaluated on transaction networks containing as many as approximately 1.1 billion nodes and 3.7 billion edges. This highlights both the practical potential of GNN-based fraud detection and the importance of scalability when applying graph-learning techniques to real-world financial systems.

SEFraud

Li et al. proposed **SEFraud**, a graph-based self-explainable fraud detection framework. In contrast to post-hoc explanation techniques such as GNNExplainer, SEFraud incorporates interpretability directly into the fraud detection model.

The authors observed that post-hoc methods introduce additional computational cost and do not directly contribute to the prediction mechanism of the fraud detector. To address this issue, SEFraud uses a customized heterogeneous graph transformer combined with learnable feature masks and edge masks. These masks identify important node

characteristics and graph relationships while the fraud detection model itself is being trained.

SEFraud also introduces a contrastive triplet loss intended to strengthen the learning of meaningful explanatory masks. In this way, both detection and interpretation are optimized within a unified framework. The authors report improvements in fraud detection performance as well as faster explanation generation compared with post-hoc approaches.

An important aspect of SEFraud is its real-world application. The framework was deployed within the Industrial and Commercial Bank of China, where the generated explanations were reported to correspond with expert business understanding. This demonstrates that explainability can serve a practical function in financial fraud detection rather than being limited to academic visualization.

Comparative Analysis and Research Motivation

The reviewed literature demonstrates considerable progress in both Graph Neural Network-based fraud detection and graph explainability. GNN architectures provide mechanisms for capturing relational patterns that cannot easily be represented using conventional independent transaction features. At the same time, explainability techniques attempt to make the increasingly complex decisions of these models understandable.

GNNExplainer provides detailed instance-level explanations using graph and feature masks, but repeated optimization can become computationally expensive. PGExplainer improves scalability by learning a parameterized explanation model, although its explanations are primarily focused on graph edges. xFraud demonstrates how GNN-based detection and explanation can be integrated in a large-scale real-world transaction system, while SEFraud takes a further step by incorporating explainability directly into the detection model.

These studies also indicate that there is no single GNN architecture or explanation approach that is universally optimal for every financial fraud detection scenario. Different models provide different capabilities in terms of representation learning, complexity, scalability, and interpretability. Furthermore, the recent review of financial fraud detection using GNNs identifies improved explainability as an important direction for future research and emphasizes the need for techniques that clarify the significance of nodes, edges, and other components involved in GNN decisions.

Based on these observations, the present study adopts a **comparative approach** to investigate explainable financial fraud detection using Graph Neural Networks. Different GNN-based models will be evaluated for their fraud classification performance, while GNNExplainer will be used to examine the reasoning behind individual predictions. The comparison is intended to determine how architectural differences affect fraud detection performance while simultaneously demonstrating how explainability can provide additional insight into graph-based predictions.

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